NVIDIA RTX A6000 — Hierarchical Roofline
FP16 GEMM: 3072 × 3072 × 3072

Performance [GFLOP/s]

10^5
10^4
10^3

L1: 9,104.3 GB/s
L2: 2,409.6 GB/s
DRAM (GDDR6): 710.0 GB/s
Measured dense FP16 Tensor: 160.44 TFLOP/s
Measured FP32 (ERT): 24.81 TFLOP/s
Theoretical FP32: 38.7 TFLOP/s
Theoretical dense FP16 Tensor: 154.8 TFLOP/s
Theoretical GDDR6: 768 GB/s

GEMM implementation
v1
v2
v3
v4
v5
v6
v7
v8
cublas
cutlass

10^1
10^2
10^3

Arithmetic intensity [effective FLOP / measured byte]

Points use matched kernel duration and interface traffic. Lines connect the same implementation.
Solid: measured. Dashed: whitepaper Table 3 (1800 MHz). Dense GEMMs; no 2:4 sparsity.
Shared GPU workload: empirical roofs are observed values, not isolated hardware peaks.